PROJECT PHASE BREAKDOWN

Hybrid Physics-ML Modeling of Fluoride Adsorption on Coconut Husk

Project Duration: 3-4 weeks
Total Effort: ~80-100 hours
Status: Phase 1 Research Complete → Phase 1 Execution Starting

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EXECUTIVE SUMMARY

This document outlines all **8 phases** of the fluoride adsorption hybrid modeling project:

Phase	Name	Status	Duration	Deliverables
1	Research + DoE	**IN PROGRESS**	Week 1	Literature review, DoE matrix, dataset
2	Langmuir Fitting	Queued	Week 2	Chemical model, isotherms, R^2 values
3	Residual Analysis	Queued	Week 2	ML features, pattern detection
4	ML Training	Queued	Week 2-3	RF/XGBoost/MLP models, cross-validation
5	Hybrid Integration	Queued	Week 3	Combined model, comparison report
6	Dashboard Dev	Queued	Week 3	Streamlit app, interactive UI
7	Visualization	Queued	Week 3-4	Figures, plots, 3D surfaces
8	Documentation	Queued	Week 4	Final report, paper-ready version

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PHASE-BY-PHASE DETAILS

PHASE 1: RESEARCH & DATA GENERATION

Status: IN PROGRESS
Duration: 1 week (3 days complete, 4 days remaining)
Location: Week 1

1.1: Literature Foundation COMPLETE Deliverable: `phase1_research_report.md` (10,000+ words) **Includes:** - Langmuir 1918 original paper analysis - 35+ peer-reviewed studies on fluoride adsorption - Coconut husk specific research - Modified/engineered adsorbent benchmarks - Hybrid modeling approaches - DoE/RSM validation

Key Findings: - Coconut husk q_max: 6.5-8.5 mg/g (unmodified), up to 54.48 mg/g (modified) - Optimal pH: 5-7 (validated from 15+ studies) - Kinetics: Pseudo-second-order (PSO) in all cases - Isotherms: Langmuir (R^2 0.85-0.99)

1.2: DoE Matrix Generation IN PROGRESS Deliverable: `data/doe_matrix.csv` **Specification:** - Design Type: Face-Centered Central Composite Design (CCD) - Factors: 5 (pH, C , Time, Temp, Flow) - Runs: 48 total - Factorial points: 32 - Axial points: 10 - Center replicates: 6 - Ranges (literature-validated): `pH: 3.0 - 9.0 (center: 6.0) C : 1.0 - 10.0 mg/L (center: 5.5) Time: 10 - 120 min (center: 65) Temp: 20 - 40 °C (center: 30) Flow: 0.5 - 2.0 L/min (center: 1.25)`

Output: `data/doe_matrix.csv` - 48 rows × 8 columns - Columns: Run, pH, C0, Time, Temp, Flow, Order, (metadata) - Ready for simulation

Scripts: `src/doe_design.py`

1.3: Physics-Based Simulation IN PROGRESS Deliverable: `data/dataset_simulated.csv` **Simulation Components:** 1. Langmuir equilibrium (T-dependent) 2. pH effect (Gaussian bell curve, peak at 6.5) 3. Pseudo-2nd-order kinetics 4. Temperature correction (Arrhenius, E_a=20 kJ/mol) 5. Flow rate effect (residence time) 6. Realistic noise (±5% analytical error)

Physical Parameters (Literature-Validated):

```
q_max = 8.5 mg/g (Talat et al. 2018)
K_L = 0.12 L/mg @ 25°C
E_a = 20 kJ/mol
pH_opt = 6.5
pH_  = 1.5
k  = 0.05 g/(mg·min)
Noise = ±5%
```

Output: `data/dataset_simulated.csv` - 48 rows × 7 columns - Columns: Run, pH, C0_mg_L, Time_min, Temp_C, Flow_L_min, Efficiency_percent, Capacity_mg_g - Value ranges: - Efficiency: 15-97% - Capacity: 0.5-8.5 mg/g

Scripts: `src/simulate_adsorption.py`

1.4: Data Validation IN PROGRESS Deliverable: `output/01_data_exploration.png` **Validation Checks:** - [ ] pH shows bell curve (peak 6.5) - [ ] Time shows saturation curve (>90% @ 120 min) - [ ] Temperature shows positive effect (+5-10% per 20°C) - [ ] Flow rate shows negative effect (-~10% per 0.5 L/min) - [ ] Distribution: mean ~71%, std ~22% - [ ] No negative values, NaNs, or anomalies

Plots (6 subplots): 1. pH vs Efficiency (bell curve) 2. Time vs Efficiency (saturation) 3. Temperature vs Efficiency (positive) 4. Concentration vs Efficiency 5. Flow Rate vs Efficiency (negative) 6. Distribution histogram

Scripts: `notebooks/01_data_exploration.py`

1.5: Deliverables Summary

File	Type	Size	Status
`phase1_research_report.md`	Markdown	~50 KB	Complete
`phase1_quick_reference.md`	Markdown	~20 KB	Complete
`doe_matrix.csv`	CSV	~5 KB	Ready

File	Type	Size	Status
`dataset_simulated.csv`	CSV	~8 KB	Ready
`01_data_exploration.png`	PNG	~200 KB	Ready

PHASE 2: CHEMICAL MODEL FITTING

Status: QUEUED (starts after Phase 1.5)
Duration: 2 days
Estimated Effort: 6-8 hours

Objectives

1. Fit Langmuir isotherm (single and dual-site)
2. Extract model parameters (qmax, KL)
3. Calculate statistical metrics (R^2, RMSE, residuals)
4. Validate model assumptions

2.1: Single-Site Langmuir (SSL) Fitting **Model:** q_e = (qmax × KL × C_e) / (1 + KL × C_e)

Process: 1. Prepare data (equilibrium qe, Ce pairs) 2. Non-linear curve fitting (scipy.optimize.curve_fit) 3. Parameter extraction: qmax, KL, confidence intervals 4. Statistical analysis: R^2, RMSE, AIC, BIC 5. Residual analysis

Expected Results: - q_max: 8.0-9.0 mg/g - KL: 0.10-0.15 L/mg - R^2: 0.88

Deliverable: `models/langmuir_ssl.pkl`, `reports/ssl_fit_report.pdf`

2.2: Dual-Site Langmuir (DSL) Testing **Model:** q_e = (q1 × KL1 × C_e)/(1 + KL1 × C_e) + (q2 × KL2 × C_e)/(1 + KL2 × C_e)

Process: 1. Fit two-site model 2. Compare with SSL (F-test) 3. Determine if heterogeneity is significant 4. Report: q1, q2, KL1, KL2

Expected Results: - Site 1 (high-energy): q1 5-6 mg/g, KL1 0.2-0.3 L/mg - Site 2 (low-energy): q2 2-3 mg/g, KL2 0.05-0.08 L/mg - R^2: 0.92 (improvement 3-5% vs SSL)

Deliverable: `models/langmuir_dsl.pkl`, `reports/dsl_fit_report.pdf`

2.3: Temperature Dependence **Analysis:** 1. Extract KL vs T data 2. Fit Arrhenius equation: KL(T) = KL_ref × exp[(E_a/R)(1/T_ref - 1/T)] 3. Extract E_a (activation energy) 4. Thermodynamic analysis: $\Delta G°$, $\Delta H°$, $\Delta S°$

Expected Results: - E_a: 15-25 kJ/mol (endothermic) - $\Delta G°$: -40 to -80 kJ/mol (spontaneous) - Temperature correction: ~7% increase from 20°C to 40°C

Deliverable: `reports/thermodynamic_analysis.pdf`

2.4: Kinetic Modeling **Models to fit:** 1. Pseudo-first-order (PFO) 2. Pseudo-second-order (PSO) 3. Intra-particle diffusion (IPD)

Analysis: - Plot qt vs t data - Fit three models - Compare R^2 values - Select best model

Expected Results: - PSO best fit (R^2 0.98) - k : 0.03-0.07 g/(mg · min) - Eq. reached in 30-60 min

Deliverable: `reports/kinetic_analysis.pdf`

2.5: Phase 2 Deliverables

Deliverable	Format	Purpose
`ssl_fit_report.pdf`	PDF	Parameter values, R^2, analysis
`dsl_fit_report.pdf`	PDF	Dual-site model results
`thermodynamic_analysis.pdf`	PDF	E_a, ΔH, ΔS, ΔG
`kinetic_analysis.pdf`	PDF	PSO/PFO/IPD fitting
`langmuir_ssl.pkl`	Pickle	Fitted model object
`langmuir_dsl.pkl`	Pickle	Fitted model object
`02_langmuir_comparison.png`	PNG	Isotherm plots, all factors

PHASE 3: RESIDUAL ANALYSIS

Status: QUEUED
Duration: 1.5 days
Estimated Effort: 4-5 hours

Objectives

1. Analyze residuals from chemical model
2. Detect systematic patterns
3. Identify ML features
4. Prepare for ML correction layer

3.1: Residual Diagnostics **Analysis:** 1. Plot residuals vs fitted values (Bland-Altman plot) 2. Q-Q plot (normality check, Shapiro-Wilk test) 3. Autocorrelation (Durbin-Watson, ACF plots) 4. Residuals vs each predictor

Expected Findings: - Non-zero mean residuals at pH extremes - Heteroscedasticity at high capacity - Correlation with time (kinetic lag) - Correlation with flow rate

3.2: Feature Engineering **Physics-Informed Features to Create:** 1. `pH_squared` = pH^2 2. `pH_deviation` = $(pH - 6.5)^2$ 3. `log_time` = log(Time) 4. `time_squared` = $Time^2$ 5. `normalized_temp` = (Temp - 25) / 10 6. `residence_time` = Time / Flow 7. `log_concentration` = log(C0) 8. `pH_x_time_interaction` = pH × log(Time) 9. `temp_x_concentration` = Temp × C0 10. `capacity_predicted` = q_e from chemical model

Rationale: These features capture mechanisms not in simple Langmuir model

3.3: Feature Selection **Method:** RandomForestRegressor for initial importance ranking **Select:** Top 8-10 features with |importance| > 0.01

Deliverable: `features_selected.csv`

```
Feature,Importance,Selected
pH_deviation,0.25,Yes
log_time,0.18,Yes
residence_time,0.15,Yes
normalized_temp,0.12,Yes
...
```

3.4: Phase 3 Deliverables

Deliverable	Format	Purpose
`residual_diagnostics.pdf`	PDF	All diagnostic plots
`features_engineered.csv`	CSV	All 10 engineered features
`features_selected.csv`	CSV	Top 8 features for ML
`03_feature_importance.png`	PNG	Bar plot of importances

PHASE 4: MACHINE LEARNING MODELS

Status: QUEUED
Duration: 2-3 days
Estimated Effort: 10-12 hours

Objectives

1. Train ML models on residuals
2. Cross-validate all models
3. Select best performing model
4. Prepare for hybrid integration

4.1: ML Feature Preparation **Features:** 8 selected physics-informed features **Target:** Residuals from chemical model (q_pred_chemical) **Split:** 70% train, 30% test (with stratification on pH bins) **Scaling:** StandardScaler on all features

Data Summary: - Training samples: 34 - Test samples: 14 - Features: 8 - Target range: ±2 mg/g (typical residual magnitude)

4.2: Models to Train

1. **Random Forest (RF)**
   - n_estimators: 50-100
   - max_depth: 3-5
   - min_samples_leaf: 3
   - GridSearchCV for hyperparameters

2. **XGBoost (XGB)**
    - max_depth: 3-4
    - learning_rate: 0.01-0.1
    - n_estimators: 50-200
    - GridSearchCV
3. **Multi-Layer Perceptron (MLP)**
    - Architecture: [8, 32, 16, 8, 1]
    - Activation: ReLU (hidden), Linear (output)
    - Regularization: L2 (=0.001)
    - Early stopping (patience=10)

4.3: Cross-Validation Strategy **Method:** 5-fold stratified cross-validation **Metrics:** - R^2 (coefficient of determination) - RMSE (root mean squared error) - MAE (mean absolute error) - AARD% (absolute average relative deviation)

Expected Results:

```
Model       R²      RMSE    MAE     AARD%
RF          0.92    0.45    0.32    4.2%
XGBoost     0.90    0.48    0.35    4.8%
MLP         0.88    0.52    0.38    5.1%
Ensemble    0.93    0.42    0.30    3.8%
```

4.4: Hyperparameter Tuning **Grid Search for RF:**

```
param_grid = {
    'n_estimators': [50, 100, 150],
    'max_depth': [3, 4, 5],
    'min_samples_leaf': [2, 3, 5],
}
GridSearchCV(RF, param_grid, cv=5, scoring='r2')
```

Result: Best RF configuration saved

4.5: Feature Importance Analysis **For each model:** 1. Calculate feature importance scores 2. Plot top features 3. Interpretation: which mechanisms matter most?

Expected Top Features: 1. pH_deviation (30-40%) 2. log_time (20-25%) 3. residence_time (15-20%) 4. normalized_temp (10-15%)

4.6: Phase 4 Deliverables

Deliverable	Format	Purpose
`ml_models.pkl`	Pickle	All trained models (RF, XGB, MLP)
`ml_cv_results.csv`	CSV	Cross-validation scores per fold
`ml_best_hyperparams.json`	JSON	Best parameters for each model
`04_model_comparison.png`	PNG	Box plots: R^2, RMSE comparison
`04_feature_importance_ml.png`	PNG	Feature importance per model
`ml_model_report.pdf`	PDF	Full training report

PHASE 5: HYBRID MODEL INTEGRATION

Status: QUEUED
Duration: 1.5 days
Estimated Effort: 6-7 hours

Objectives

1. Integrate chemical + ML models
2. Compare against pure approaches
3. Validate hybrid superiority
4. Prepare for deployment

5.1: Hybrid Architecture

```
INPUT (5 factors)
    ↓
[Chemical Layer: Langmuir SSL]
    ↓
q_chemical, residual_estimate
    ↓
[ML Layer: Best Model (RF/XGB)]
    ↓
residual_correction
    ↓
q_final = q_chemical + residual_correction
    ↓
OUTPUT: Final Capacity Prediction
```

5.2: Model Comparison **Three Approaches:** 1. **Chemical-Only (Baseline)** - q_pred = Langmuir(pH, C0, T) - R^2: ~0.85-0.88 - RMSE: ~0.9-1.2 mg/g

2. **ML-Only (Black Box)**
    - q_pred = RF(all 8 features)
    - R^2: ~0.92-0.95
    - RMSE: ~0.45-0.55 mg/g
    - Issue: Not interpretable
3. **Hybrid (Physics+ML)** PROPOSED
    - q_pred = Langmuir(…) + RF(residuals)
    - Expected R^2: ~0.93-0.96
    - Expected RMSE: ~0.4-0.5 mg/g
    - Benefit: Interpretable + accurate

5.3: Validation on Test Set **Test Set:** 14 samples (held out from Phase 4)

Evaluation:

```python
y_true = actual capacities
y_chem = chemical model predictions
y_ml = ML model predictions
y_hybrid = hybrid model predictions

# Metrics for each approach
for y_pred in [y_chem, y_ml, y_hybrid]:
    R² = r2_score(y_true, y_pred)
    RMSE = np.sqrt(mean_squared_error(y_true, y_pred))
    MAE = mean_absolute_error(y_true, y_pred)
    print(f"R²: {R²:.4f}, RMSE: {RMSE:.3f}, MAE: {MAE:.3f}")
```

Expected Results: - Chemical: R^2 0.86, RMSE 1.05 - ML: R^2 0.93, RMSE 0.48 - Hybrid: R^2 0.94, RMSE 0.43 BEST

5.4: Error Analysis **Residual Plots:** 1. Predicted vs Actual (scatter + 1:1 line) 2. Residuals vs Predicted (should be random) 3. Residuals histogram (should be normal) 4. Residuals vs pH, Time, Temp, Flow (should be random)

Outlier Detection: - Identify >±0.5 mg/g errors - Investigate causes - Document limitations

5.5: Uncertainty Quantification **For Hybrid Model:** 1. Prediction intervals (±95% CI) 2. Based on residual standard deviation 3. Account for both chemistry and ML uncertainty

Formula:

```
q_pred ± 1.96 ×  _residuals
```

5.6: Phase 5 Deliverables

Deliverable	Format	Purpose
`hybrid_model.pkl`	Pickle	Combined chemical + ML model
`model_comparison_report.pdf`	PDF	All three approaches compared
`05_predictions_vs_actual.png`	PNG	Parity plots for all three
`05_residual_analysis.png`	PNG	Residual diagnostics (4 plots)
`validation_metrics.csv`	CSV	R^2, RMSE, MAE for all models

PHASE 6: STREAMLIT DASHBOARD

Status: QUEUED
Duration: 1.5 days
Estimated Effort: 5-6 hours

Objectives

1. Build interactive web interface
2. Allow real-time predictions

3. Visualize model behavior
4. Enable stakeholder exploration

6.1: Dashboard Features **1. Prediction Simulator** - Input sliders: pH, C0, Time, Temp, Flow - Real-time output: Efficiency %, Capacity mg/g - Three predictions: Chemical, ML, Hybrid - Confidence intervals

2. Model Comparison - Side-by-side R^2 scores - RMSE comparison - Model uncertainty quantification - Feature importance plots

3. Data Visualization - DoE matrix heatmap - Response surface plots (3D) - Contour plots (2D slices) - Pareto front (Efficiency vs Capacity)

4. Literature Benchmark - Compare with published capacities - Show where coconut husk ranks - Table of adsorbents

5. Model Insights - Feature importance (bar chart) - Residual diagnostics - Cross-validation curves - Model training history (MLP)

6.2: Technical Stack

```
Frontend: Streamlit
Backend: Python (scikit-learn, XGBoost)
Data: pandas, numpy
Viz: plotly, matplotlib, seaborn
Server: localhost (development), AWS/Heroku (production)
```

```python
# app.py
import streamlit as st
from src.models import load_hybrid_model
from src.simulation import FluorideAdsorptionSimulator

st.title("Fluoride Adsorption Hybrid Model")

# Sidebar: Input controls
with st.sidebar:
    pH = st.slider("pH", 3, 9, 6)
    C0 = st.slider("Initial Concentration (mg/L)", 1, 10, 5)
    Time = st.slider("Contact Time (min)", 10, 120, 65)
    Temp = st.slider("Temperature (°C)", 20, 40, 30)
    Flow = st.slider("Flow Rate (L/min)", 0.5, 2.0, 1.25)

# Main: Predictions
model = load_hybrid_model()
q_chem, q_ml, q_hybrid = model.predict(pH, C0, Time, Temp, Flow)

col1, col2, col3 = st.columns(3)
col1.metric("Chemical Model", f"{q_chem:.2f} mg/g", f"R²={0.87:.3f}")
col2.metric("ML Model", f"{q_ml:.2f} mg/g", f"R²={0.93:.3f}")
```

```python
col3.metric("Hybrid Model", f"{q_hybrid:.2f} mg/g", f"R²={0.94:.3f}")

# Tabs: Different views
tab1, tab2, tab3 = st.tabs(["Comparison", "Visualizations", "Insights"])

with tab1:
    # Model comparison plots

with tab2:
    # Response surface plots

with tab3:
    # Feature importance, residuals, etc.
```

6.3: App Structure

6.4: Deployment Options **Local Development:**

```
streamlit run app.py
# Access at http://localhost:8501
```

Cloud Deployment:

```bash
# Option 1: Heroku
git push heroku main

# Option 2: AWS EC2
aws ec2 run-instances --image-id ami-xxx --instance-type t2.micro

# Option 3: Streamlit Cloud (easiest)
# Push to GitHub, connect to Streamlit Cloud
```

6.5: Phase 6 Deliverables

Deliverable	Format	Purpose
`app/app.py`	Python	Main Streamlit app
`app/requirements.txt`	TXT	App dependencies
`app/.streamlit/config.toml`	TOML	Streamlit config
`app/README.md`	Markdown	How to run app
Dashboard screenshots	PNG	Documentation

PHASE 7: ADVANCED VISUALIZATIONS

Status: QUEUED
Duration: 1.5 days
Estimated Effort: 4-5 hours

Objectives

1. Create publication-quality figures
2. 3D surface plots
3. Interactive visualizations
4. Final data story

7.1: 3D Response Surface Plots Type 1: pH vs Time (at center values of C0, Temp, Flow)

```
Axes: pH (3-9), Time (10-120), Efficiency (%)
Shows: Bell curve in pH, saturation in time
```

Type 2: Temp vs Flow (at center values of pH, C0, Time)

```
Axes: Temperature (20-40), Flow (0.5-2.0), Capacity (mg/g)
Shows: Temperature increase, flow decrease
```

Type 3: C0 vs pH (at center values of Time, Temp, Flow)

```
Axes: Concentration (1-10), pH (3-9), Efficiency (%)
Shows: Concentration effect combined with pH
```

7.2: Contour Plots (2D Slices) For each 3D plot, create contour version - Shows iso-efficiency/capacity lines - Easier to read than 3D - Include gradient arrows

7.3: Comparison Visualizations Multi-Model Predictions:

```
For a grid of (pH, Time) at fixed C0, Temp, Flow:
- Plot 1: Chemical model surface
- Plot 2: ML model surface
- Plot 3: Hybrid model surface
- Plot 4: Difference (Hybrid - Chemical)
```

Shows where ML correction is most important (likely pH extremes)

```python
import plotly.graph_objects as go

# 3D scatter plot: Actual vs Predicted
fig = go.Figure(data=[
    go.Scatter3d(x=pH, y=Capacity, z=Efficiency, mode='markers',
                 name='Actual', marker=dict(size=5, color='blue')),
    go.Scatter3d(x=pH_pred, y=Capacity_pred, z=Efficiency_pred,
                 mode='markers', name='Predicted',
                 marker=dict(size=5, color='red'))
])
fig.show()
```

7.4: Interactive Plots (Plotly)

7.5: Publication-Quality Figures **Figure 1: Langmuir Isotherm Fits** - SSL vs DSL vs actual data - Error bars on data - R^2 legends

Figure 2: pH Bell Curve - Data points - Gaussian fit - Width ( ) annotation

Figure 3: Pseudo-2nd-Order Kinetics - Time course - PSO model fit - Equilibrium line

Figure 4: Temperature Effect - Arrhenius plot (ln(KL) vs 1/T) - E_a extraction - Confidence bands

Figure 5: Model Comparison - Three models side-by-side - R^2 and RMSE bar charts - Parity plots

Figure 6: Feature Importance - Horizontal bar chart - Sorted by importance - Color by category (pH, kinetic, thermal, etc.)

Figure 7: 3D Response Surface - Best visualization - All angles - Color gradient for Z-axis

Figure 8: Hybrid Model Error Distribution - Histogram of residuals - Normal distribution overlay - Outliers marked

7.6: Phase 7 Deliverables

Deliverable	Format	Purpose
`figures/01_langmuir_fits.pdf`	PDF	Isotherm comparison
`figures/02_pH_bell_curve.pdf`	PDF	pH dependence
`figures/03_kinetics.pdf`	PDF	PSO fitting
`figures/04_arrhenius.pdf`	PDF	Temperature effect
`figures/05_model_comparison.pdf`	PDF	All models compared
`figures/06_feature_importance.pdf`	PDF	ML feature ranks
`figures/07_3d_surface.html`	HTML	Interactive 3D
`figures/08_residuals.pdf`	PDF	Error analysis

PHASE 8: DOCUMENTATION & REPORTING

Status: QUEUED
Duration: 2 days
Estimated Effort: 8-10 hours

Objectives

1. Write comprehensive final report
2. Create publication-ready manuscript
3. Document all methods and results
4. Provide reproducibility guidelines

8.1: Final Report Structure

```
FINAL_REPORT.md / FINAL_REPORT.pdf
  Executive Summary
  1. Introduction
     Fluoride contamination problem
     Adsorption treatment overview
     Hybrid modeling approach
  2. Literature Review (condensed)
     Langmuir theory
     Fluoride adsorption
     Coconut husk adsorbents
     ML in water treatment
  3. Methodology
     DoE Design
     Simulation physics
     Chemical model (Langmuir)
     ML models (RF, XGB, MLP)
     Hybrid integration
  4. Results
     Dataset characteristics
     Chemical model fits
     Kinetic analysis
     ML model performance
     Hybrid model validation
  5. Discussion
     Parameter interpretation
     Mechanism insights
     Comparison with literature
     Hybrid advantages
  6. Conclusions
     Key findings
     Research contributions
     Future work
  7. References
     40+ cited papers
  Appendices
      A: DoE Matrix
      B: Detailed Equations
      C: Hyperparameter Tuning
      D: Cross-Validation Results
      E: Code Snippets
```

8.2: Key Sections Detail **Executive Summary (1 page)** - Problem statement - Approach (hybrid physics-ML) - Key findings (R^2, RMSE) - Conclusion

Introduction (3-4 pages) - Fluoride health impacts - Current treatment methods - Coconut husk potential - Hybrid modeling gap - Research objectives

Methodology (5-6 pages) - DoE (CCD design, 5 factors, 48 runs) - Simulation (6 mechanisms, parameters) - Chemical model (Langmuir SSL/DSL) - ML models (RF, XGB, MLP) - Hybrid architecture - Validation approach

Results (8-10 pages) - Dataset summary (Table) - Langmuir parameters (q_max, KL, R^2) - Kinetic analysis (k , R^2, PSO vs PFO) - Temperature effects (E_a, ΔG, ΔH, ΔS) - ML model cross-validation (Table: RF, XGB, MLP R^2 RMSE) - Hybrid model test set performance - Error analysis, outliers

Discussion (5-7 pages) - Parameter interpretation vs literature - Why Langmuir insufficient - ML correction insights (feature importance) - Hybrid advantages quantified - Limitations and uncertainties - Application to real systems

Conclusions (1-2 pages) - Key contributions - Hybrid model superiority (15-30% RMSE reduction) - Practical implications - Future work (real data, other adsorbents, other contaminants)

8.3: Supplementary Materials **Table S1: Literature Parameter Compilation** - 20+ studies: qmax, KL, pH_opt - Adsorbent type comparison - Validation of parameter ranges

Table S2: DoE Matrix (all 48 runs) - Printable reference - Run order

Table S3: Dataset (all 48 runs + predicted) - Actual vs Chemical vs ML vs Hybrid - Residuals

Figure S1-S8: All Detailed Plots - Higher resolution versions

Figure S9: Model Architecture Diagram - Visual representation of hybrid system

Code Snippet S1: Langmuir Fitting

```python
# How to fit Langmuir isotherm
from scipy.optimize import curve_fit

def langmuir(Ce, qmax, KL):
    return (qmax * KL * Ce) / (1 + KL * Ce)

popt, pcov = curve_fit(langmuir, Ce_data, qe_data)
qmax, KL = popt
```

8.4: Reproducibility Package **To enable others to reproduce:** 1. `requirements.txt` - exact versions 2. `data/doe_matrix.csv` - input design 3. `data/dataset_simulated.csv` - generated data 4. `src/*.py` - all code, well-commented 5. `notebooks/*.ipynb` - step-by-step notebooks 6. `models/*.pkl` - trained models 7. `README.md` - how to run everything

README Structure:

````markdown
# Fluoride Adsorption Hybrid Model

## Quick Start
```bash
pip install -r requirements.txt
python src/simulate_adsorption.py
python src/phase2_langmuir_fitting.py
...
````
````

Project Structure

- `data/`: Datasets
- `src/`: All Python code
- `notebooks/`: Jupyter tutorials
- `models/`: Saved ML models
- `reports/`: Final outputs

Methodology

- Phase 1: DoE + Simulation
- Phase 2: Langmuir Fitting
- … etc

Results

- Hybrid $R^2 = 0.94$ (test set)
- 15-30% RMSE improvement vs chemical alone

References

- See FINAL_REPORT.md for full citations

```
#### 8.5: Phase 8 Deliverables
| Deliverable | Format | Purpose |
|-------------|--------|---------|
| `FINAL_REPORT.md` | Markdown | Complete technical report |
| `FINAL_REPORT.pdf` | PDF | Publication-ready version |
| `FINAL_REPORT_EXECUTIVE_SUMMARY.pdf` | PDF | 1-page executive brief |
| `README.md` | Markdown | Project overview & how to run |
| `REPRODUCIBILITY_GUIDE.md` | Markdown | Step-by-step reproduction |
| `requirements.txt` | TXT | Python dependencies |
| `ALL_TABLES.xlsx` | Excel | Summary of all results |
| `supplementary_materials.pdf` | PDF | Figures, code, extended methods |

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## PROJECT TIMELINE GANTT
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Week 1: PHASE 1 (Research, DoE, Simulation) Week 2: PHASE 2 PHASE 3 (Langmuir, Residuals) Week 3: PHASE 4 PHASE 5 (ML, Hybrid) Week 4: PHASE 6 PHASE 7 PHASE 8 (Dashboard, Visualizations, Report) “‘

KEY METRICS ACROSS ALL PHASES

Phase 1-2: Model Building

Metric	Target	Expected
DoE Runs	48	48
Factors	5	5
R^2 (Chemical)	0.85	0.87
RMSE (Chemical)	1.2	0.95

Phase 3-4: ML Development

Metric	Target	Expected
ML Features	8-10	10
Cross-Val Folds	5	5
R^2 (ML)	0.92	0.93
RMSE (ML)	0.5	0.48

Phase 5: Integration

Metric	Target	Expected
R^2 (Hybrid)	0.93	0.94
RMSE (Hybrid)	0.45	0.43
Improvement vs Chem	15%	25%

Phase 6-8: Deployment

Metric	Target	Expected
Dashboard Live	Yes	Yes
Figures (High-res)	8+	8+
Report Pages	20+	25
Code Reproducibility	100%	100%

SKILLS & KNOWLEDGE GAINED

Technical Skills

- Design of Experiments (DoE)
- Central Composite Design (CCD)
- Physics-based simulation
- Non-linear regression (curve fitting)
- Pseudo-second-order kinetics
- Arrhenius rate equations
- Hybrid physics-ML integration
- Cross-validation & hyperparameter tuning
- Ensemble methods (RF, XGBoost)

- Neural networks (MLP)
- Streamlit web development
- Publication-quality figure creation

Domain Knowledge

- Adsorption isotherm theory
- Langmuir and Freundlich models
- Fluoride contamination
- Biomass-derived adsorbents
- Water treatment engineering
- Sustainability & green chemistry

Professional Skills

- Scientific report writing
- Project management
- Reproducible research
- Version control (git)
- Code documentation
- Data visualization

ESTIMATED WORKLOAD

Phase	Duration	Effort	Notes
1	1 week	20 hrs	Heavy research + data gen
2	2 days	6 hrs	Model fitting
3	1.5 days	4 hrs	Feature engineering
4	2-3 days	10 hrs	ML training + tuning
5	1.5 days	6 hrs	Integration
6	1.5 days	5 hrs	Dashboard
7	1.5 days	4 hrs	Visualizations
8	2 days	8 hrs	Report writing
Total	**3-4 weeks**	**~85 hrs**	**Full project**

SUCCESS CRITERIA (Final)

By End of Phase 8, Project is Complete When:

1. **Literature Foundation**
   - 40+ papers reviewed
   - All mechanisms documented
   - Parameters validated
2. **Data Generated**
   - 48-run CCD implemented

- Physics-based simulation
- Dataset validated

3. **Chemical Model**
   - Langmuir SSL fitted (R^2 0.85)
   - DSL tested (R^2 0.92)
   - Parameters extracted
4. **ML Models**
   - 3 models trained (RF, XGB, MLP)
   - Cross-validated (5-fold)
   - Best model: R^2 0.92
5. **Hybrid Model**
   - Chemical + ML integrated
   - Test set R^2 0.93
   - RMSE 0.5 mg/g
   - 15-30% improvement vs chemical
6. **Dashboard**
   - Interactive prediction interface
   - Model comparison view
   - Deployable
7. **Visualizations**
   - 8+ publication-quality figures
   - 3D response surfaces
   - Interactive plots
8. **Documentation**
   - 25-page final report
   - Code fully reproducible
   - Supplementary materials
9. **Presentation**
   - Executive summary
   - Key findings highlighted
   - Insights communicated

NEXT IMMEDIATE STEPS

1. Complete Phase 1.1 (Literature) - **DONE**
2. Complete Phase 1.2-1.5 (DoE + Simulation) - **TODAY/TOMORROW**
3. Move to Phase 2 (Langmuir Fitting) - **Next 2 days**
4. Continue through Phase 8 systematically

Document Prepared: May 3, 2026
Project Status: Phase 1 Complete, Ready for Execution
Estimated Completion: Late May 2026

Ready to proceed to Phase 1 Execution!